

Curriculum and Lab Setup

What is installed and what is taught

The lab

Twenty workstations, each with a microcontroller development board, a motor and sensor kit, a small robotic arm and a shared vision rig. The equipment stays Kalpana property and is insured by Kalpana.

Semester one

Embedded C, microcontroller peripherals, sensor interfacing, motor control and serial protocols. Assessed by a working build, not by a written paper.

Semester two

Robot kinematics, PID control, computer vision basics, industrial communication and a capstone build specified by an employer partner.

Why it is written from job descriptions

The curriculum is rebuilt every year from the actual postings our 46 employer partners publish. In 2026 that added CAN bus and battery management, because every EV supplier in Hosur was asking for it and no syllabus in the state covered it.

Assessment

Every student ends with a portfolio of six working builds and a capstone. That portfolio, not the marksheet, is what goes to the employer.